

Immersive Pedigree Graph Visualisations

Septian Razi*
Australian National University

Henry Gardner†
Australian National University

Matt Adcock‡
CSIRO's Data61

ABSTRACT

We describe a study of six visualisation methods for hierarchical pedigree data in virtual reality. We model six different pedigree graph layouts: planar, cylinder, floor, sphere, cone (force directed) and vase layouts. Measurements of task accuracy and task completion time showed a statistically significant pairwise comparison on two task completion times between the sphere (better) and floor (worse) conditions. Likert ratings of user sentiments showed a statistically significant main effect of graph condition ratings of the understandability of the data with the sphere and vase graph layouts to be generally higher and the floor layout to be generally lower.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Graph drawings; Human-centered computing—Visualization—Visualization application domains—Scientific visualization; Human-centered computing—Human-computer interaction (HCI)—Interaction paradigms—Virtual reality.

1 INTRODUCTION

Pedigree graphs are hierarchical graphs that model the hierarchical offspring-parent relationship between two varieties within a species. When used to show data relating to agricultural data, some novel aspects of pedigree graphs that need to be considered for visualisation are

- *Multiple parents*, the existence of two or more parents given a single node
- *Inbreeding*, children nodes are often crossed with their parents to create new children
- *Large cross-generational crosses*, children with parents from vastly different hierarchical generations

In dealing with these considerations, common graph visualisation techniques may not be sufficient to model these aspects appropriately.

It has been shown that certain tasks in graph visualisation workflows can be performed better in an immersive environment compared with the traditional 2D desktop view [1]. The addition of an extra dimension provides another degree of freedom which is useful for highly complex data, which traditional 2D views can struggle to present well. Thus, applying these benefits towards hierarchical pedigree data might lead to improvements in their understanding.

Prior research has explored the use of Immersive Technologies with Graph Visualisations [1] [5] [6]. However, most studies in this area have used data sets that were not hierarchical in nature which means that their visualisations allowed for less constraints on the structure of these graphs (and on the tasks that would be desired to be supported by these graphs).

*e-mail: septian.razi@anu.edu.au; ORCID 0000-0003-2122-7362;

†e-mail: henry.gardner@anu.edu.au; ORCID 0000-0002-8613-8777; Henry Gardner is also a Visiting Scientist at CSIRO's Data61

‡e-mail: matt.adcock@csiro.au; ORCID 0000-0001-6191-9887; Matt Adcock is also an Adjunct Research Fellow at the Australian National University

2 RELATED WORK

There is much related work that considers simple graph visualisations in 2D; two of the most applicable graph visualisations being the force-directed method [4] and Sugiyama style [8] graphs. Conveniently, Sugiyama style graph visualisation techniques are applicable to pedigree graphs, and software suites have been developed utilising it including the Helium software package that has inspired the present work [7].

Cordeil et al have explored the use of a virtual reality headset in visualisation of quantitative data by employing embodied interaction techniques [3]. Previous work on graph visualisations with immersive technologies also exists, yet many of these studies focus on variables outside of the layout of the graph [1] [2]. Of the specific graph layouts considered here, two have been inspired by previous work. Raso et al. analysed the use of Augmented Reality technology to create a “walkable” hierarchical graph, in which nodes are laid on the ground and users are encouraged to walk around the area to understand the graph [6]. Kwon et al explored another layout, using the surface of a sphere with curved protruding edges with the user within the sphere. They found that, compared to a 2D planar view, the spherical 3D layout performed better in difficult tasks in terms of its completion time [5].

3 PROTOTYPE DESIGN

For our study, we developed a graph visualisation prototype which contained numerous features that allowed for immersive interactions with pedigree graph data, such as near and far interactions, lineage isolation, and graph manipulation. In addition to the references discussed above, many other features of our prototype were inspired by previous work related to our specific context of pedigree visualisation in immersive reality [7] [2].

Our prototype included the six different graph layout conditions from the following list (see also Figure 1):

- **Planar**, our 2D control condition
- **Cylinder**, planar that is wrapped around a user
- **Floor**, mapped to the floor radially [6]
- **Sphere**, mapped to the surface of a sphere [5]
- **Force Directed Cone**, force directed method that preserves the hierarchical levels
- **Vase**, cylindrical variation where each hierarchical level has a varying distance from the user depending on the number of nodes it contains

4 USER STUDY

4.1 Configuration

Our user study consisted of a within-subjects experiment that considered the six graph layout conditions together with six task types. We recruited 7 participants (4 male, 3 females) all of whom were undergraduate students under the age of 23 (mean = 21.14, STD = 1.24) who had access to a personal Oculus VR headset. The dataset we used for the graphs was the Golden Promise 2019 sample dataset provided by Helium [7]. However, in order to prevent learning effects for each condition we randomised the names of the data points between trials.

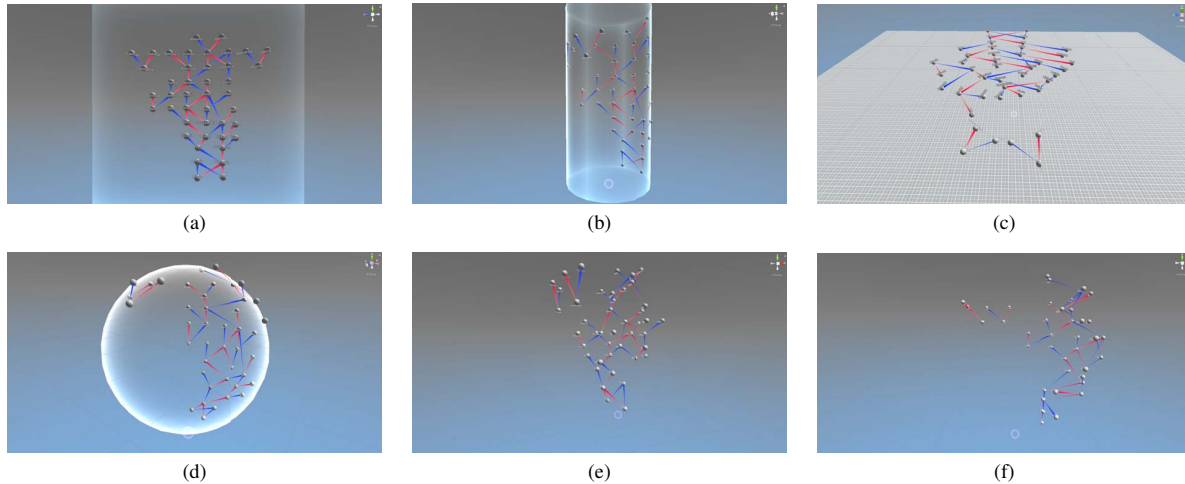


Figure 1: Examples of Graph Visualisation Layout Conditions (a) Planar (b) Cylinder (c) Floor (d) Sphere (e) Cone (Force Directed) (f) Vase

We chose to investigate tasks that are representative of a typical pedigree graph workflow. The tasks that we developed were: T1 (Find the most recent ancestors of a node), T2 (Find the most recent descendents of a node), T3 (Determine the relationship between two nodes), T4 (Find the path between two nodes), T5 (Relocate a node) and T6 (Find the Node with the largest number of children).

There were up to three tasks for each task type. We measured both the task completion time and task accuracy. After each condition we had participants fill in a questionnaire containing qualitative Likert scale questions on their experience. The experiment was approved by the Human Ethics committee of the Australian National University (protocol number 2019/738).

4.2 Results

Qualitatively, all layouts received positive feedback in terms of the questions regarding data understanding, lineage detection and experience. Overall, the sphere and vase layouts looked best across the qualitative data with the floor layout being consistently worst. We found a statistically-significant main effect of graph layout on an overall understanding of the data question according to the Kruskal-Wallis test ($H = 13.58$, $p = 0.018$). Of note, the planar graph layout performed very well in participants' ratings. This layout is essentially a 3D representation of a 2D layout and the fact that it was occasionally rated as good as or better than some 3D layouts (such as cylinder) is interesting.

Quantitatively, we found that most conditions had at least a 90% task accuracy, with the exception of the floor which received an 80% accuracy. Overall, we did not find statistically significant results using ANOVA between the quantitative metrics measured on the six graph layout conditions.

5 CONCLUSION & FUTURE WORK

Our comparison of several 3D graph layouts alongside of a 2D representative layout has not resulted in a clear winner, which may be indicative of the limited effects of layout for 3D graph visualisations. However, the sphere layout received the highest average Likert rating for overall understanding of the graph amongst all layouts.

An opportunity exists to conduct further research with these 3D graph visualisations to determine whether they are influential at all in the understanding of these graphs, or whether different variables determine this (such as the interaction techniques available or the hardware). In particular, there is a need for a even more rigorous study that may produce statistically significant results by employing a larger amount of participants.

Although pedigree graphs have unique considerations, the six structures proposed in this research can also be applied to other hierarchical graphs and may be useful for research in areas that utilise visualisations of hierarchical graph data, such as genealogy and linguistics. Tailored research in these areas could adopt the user study approach of this research, while utilising specialised interaction methods and investigating tasks representative of their unique workflows.

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